USE WIRESHARK TO ANALYZE PACKET CAPTURES AND DETECT A MAN-IN-THE-MIDDLE (MITM) ATTACK VIA ARP POISONING

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Digital Forensics – Project 2

INTRODUCTION

- Digital forensics plays a critical and key role in identifying, analyzing, and documenting cyber incidents.
- MITM attacks intercept and manipulate communication between two legitimate systems.
- ARP poisoning enables traffic redirection.
- Wireshark can be used to identify IoC`s associated with ARP poisoning.
- Traffic analysis can be conducted using endpoint statistics and detailed packet inspection.
- MITM attack can be detected using ARP traffic patterns and anomalies in MAC address mappings.

OBJECTIVE

- Analyze provided packet capture *proj_part1.pcap* file using Wireshark.
- Identify ARP poisoning activity.
- Detect evidence of MITM attack.
- Document findings using forensic methods.

ADDRESS RESOLUTION PROTOCOL (ARP)

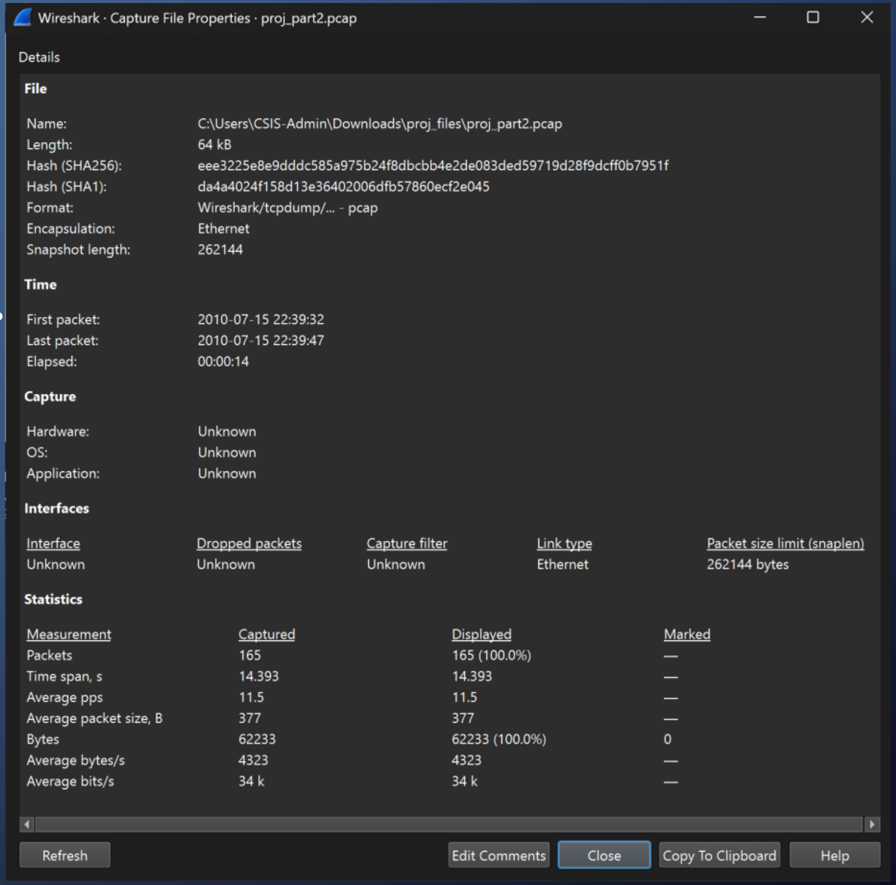
- Maps IP addresses to MAC addresses.
- Uses ARP request (broadcast).
- Uses ARP reply (unicast).
- Essential for local network communication.

MITM VIA ARP POISONING

- ARP uses request and response packets.
- Attacker sends fake ARP replies.
- Associates` attacker MAC with victim/router IP.
- Redirects traffic through attacker.
- Allows interception and modification of data.

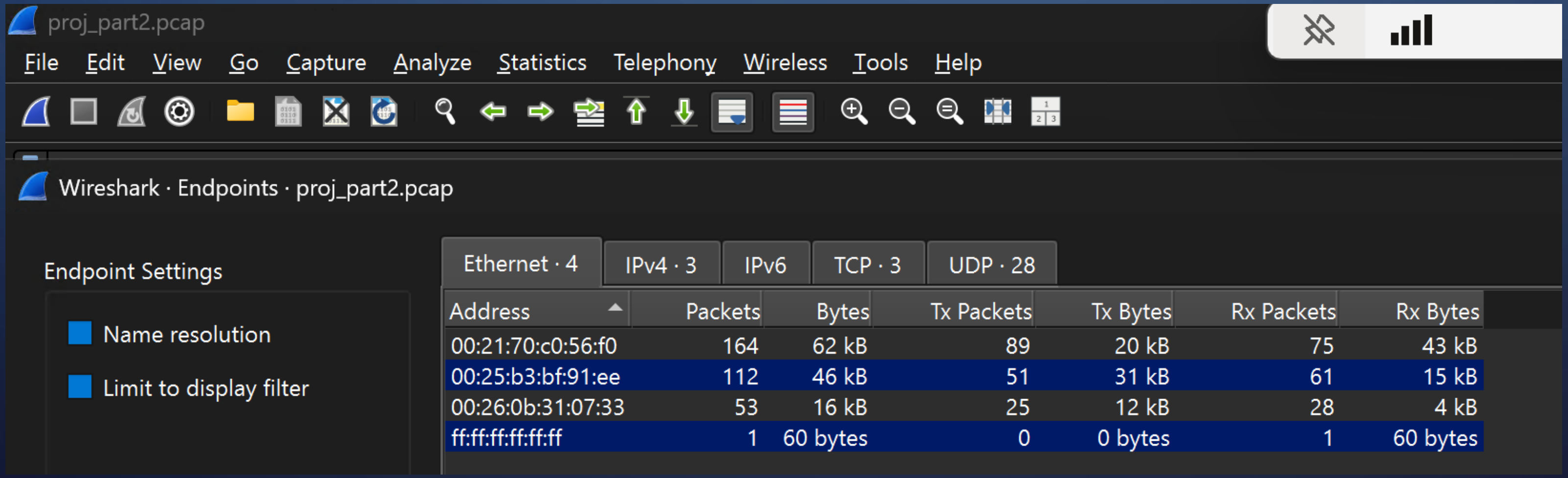
EVIDENCE VERIFICATION

- Verified file integrity using SHA256 hash.
- Ensures evidence is not altered.
- One of most important steps in digital forensics.
- Confirms reliability of analysis.



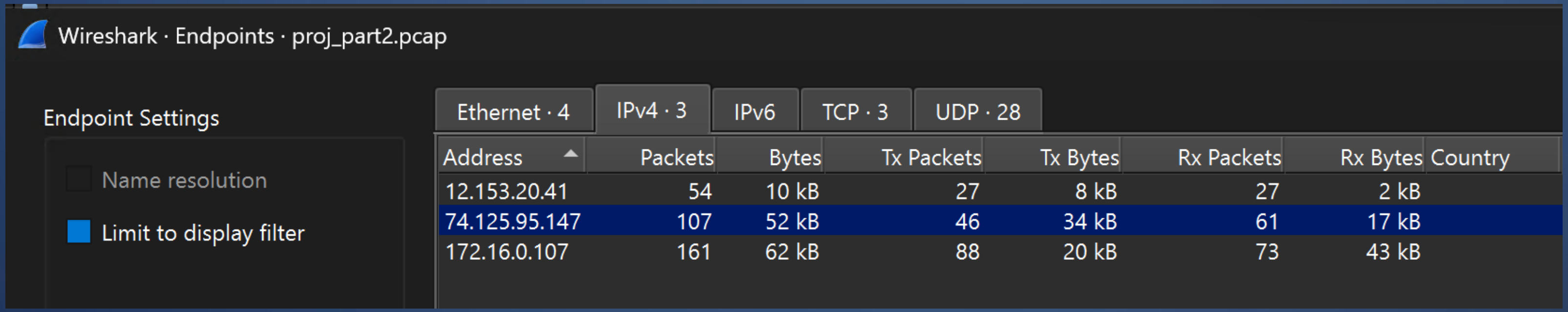
ENDPOINT ANALYSIS (ETHERNET)

- Identified victim MAC *00:21:70:c0:56:f0*
- Identified router MAC *00:26:0b:31:07:33*
- Detected attacker MAC 00:25:b3:bf:91:ee
- Observed broadcast MAC address *ff:ff:ff:ff:ff:ff*



ENDPOINT ANALYSIS (IPV4)

- Identified victim IPv4 172.16.0.107
- Identified IPv4 addresses *12.153.20.41*, and *74.125.95.147*



Address	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes	Country
12.153.20.41	54	10 kB	27	8 kB	27	2 kB	
74.125.95.147	107	52 kB	46	34 kB	61	17 kB	
172.16.0.107	161	62 kB	88	20 kB	73	43 kB	

ENDPOINT ANALYSIS (TCP)

- Displayed communication between the victim (*172.16.0.107*) and an external IP address (*74.125.95.147*) over port 80.
- Multiple packets and bytes indicate ongoing data exchange.

Wireshark · Endpoints · proj_part2.pcap

Endpoint Settings

Name resolution

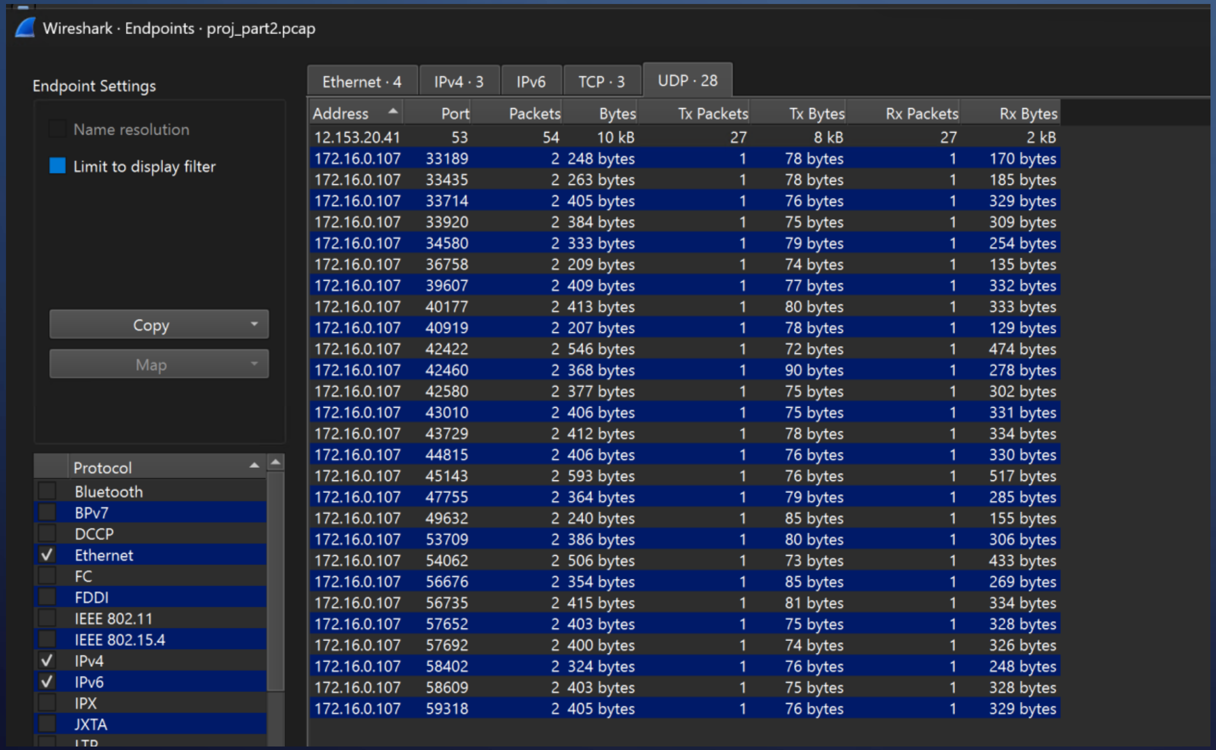
Limit to display filter

Ethernet · 4 | IPv4 · 3 | IPv6 | TCP · 3 | UDP · 28

Address	Port	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes
74.125.95.147	80	107	52 kB	46	34 kB	61	17 kB
172.16.0.107	45691	72	39 kB	40	10 kB	32	29 kB
172.16.0.107	45692	35	13 kB	21	8 kB	14	5 kB

ENDPOINT ANALYSIS (UDP)

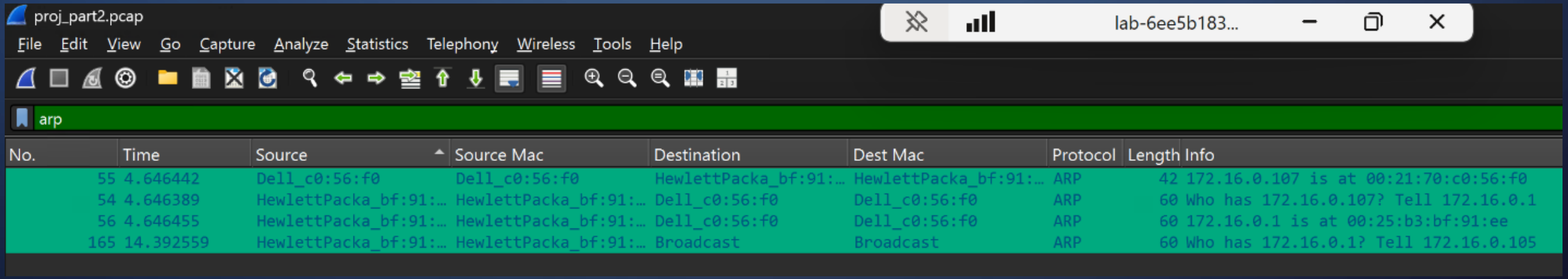
- Multiple short communications primarily involving the victim (172.16.0.107) using various ephemeral ports.
- DNS traffic over UDP (port 53) was observed.



Address	Port	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes
12.153.20.41	53	54	10 kB	27	8 kB	27	2 kB
172.16.0.107	33189	2	248 bytes	1	78 bytes	1	170 bytes
172.16.0.107	33435	2	263 bytes	1	78 bytes	1	185 bytes
172.16.0.107	33714	2	405 bytes	1	76 bytes	1	329 bytes
172.16.0.107	33920	2	384 bytes	1	75 bytes	1	309 bytes
172.16.0.107	34580	2	333 bytes	1	79 bytes	1	254 bytes
172.16.0.107	36758	2	209 bytes	1	74 bytes	1	135 bytes
172.16.0.107	39607	2	409 bytes	1	77 bytes	1	332 bytes
172.16.0.107	40177	2	413 bytes	1	80 bytes	1	333 bytes
172.16.0.107	40919	2	207 bytes	1	78 bytes	1	129 bytes
172.16.0.107	42422	2	546 bytes	1	72 bytes	1	474 bytes
172.16.0.107	42460	2	368 bytes	1	90 bytes	1	278 bytes
172.16.0.107	42580	2	377 bytes	1	75 bytes	1	302 bytes
172.16.0.107	43010	2	406 bytes	1	75 bytes	1	331 bytes
172.16.0.107	43729	2	412 bytes	1	78 bytes	1	334 bytes
172.16.0.107	44815	2	406 bytes	1	76 bytes	1	330 bytes
172.16.0.107	45143	2	593 bytes	1	76 bytes	1	517 bytes
172.16.0.107	47755	2	364 bytes	1	79 bytes	1	285 bytes
172.16.0.107	49632	2	240 bytes	1	85 bytes	1	155 bytes
172.16.0.107	53709	2	386 bytes	1	80 bytes	1	306 bytes
172.16.0.107	54062	2	506 bytes	1	73 bytes	1	433 bytes
172.16.0.107	56676	2	354 bytes	1	85 bytes	1	269 bytes
172.16.0.107	56735	2	415 bytes	1	81 bytes	1	334 bytes
172.16.0.107	57652	2	403 bytes	1	75 bytes	1	328 bytes
172.16.0.107	57692	2	400 bytes	1	74 bytes	1	326 bytes
172.16.0.107	58402	2	324 bytes	1	76 bytes	1	248 bytes
172.16.0.107	58609	2	403 bytes	1	75 bytes	1	328 bytes
172.16.0.107	59318	2	405 bytes	1	76 bytes	1	329 bytes

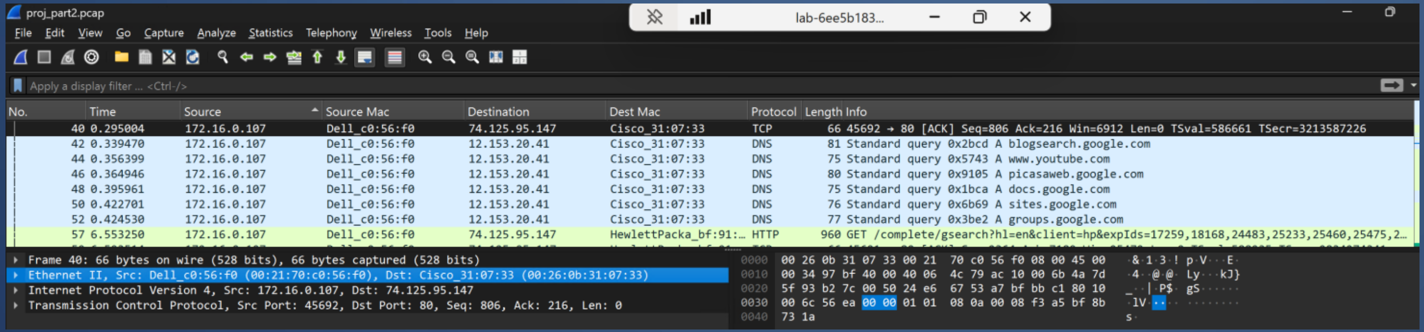
ARP IDENTIFICATION

- Used Wireshark coloring rules.
- Applied display filter arp
- Isolated ARP request and response packets
- Enabled focused ARP traffic analysis

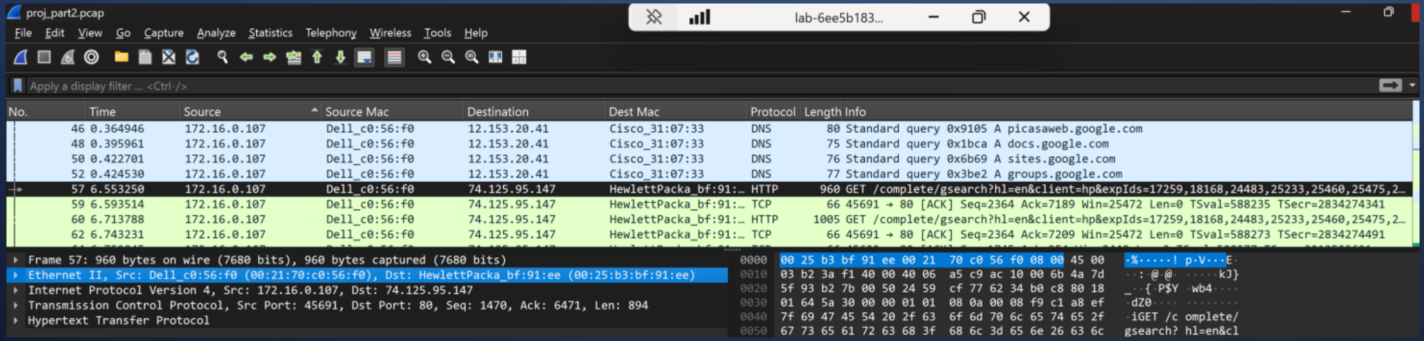


PACKET ANALYSIS (PACKET 40 AND 57)

- Packet 40 - Normal communication (victim to router)

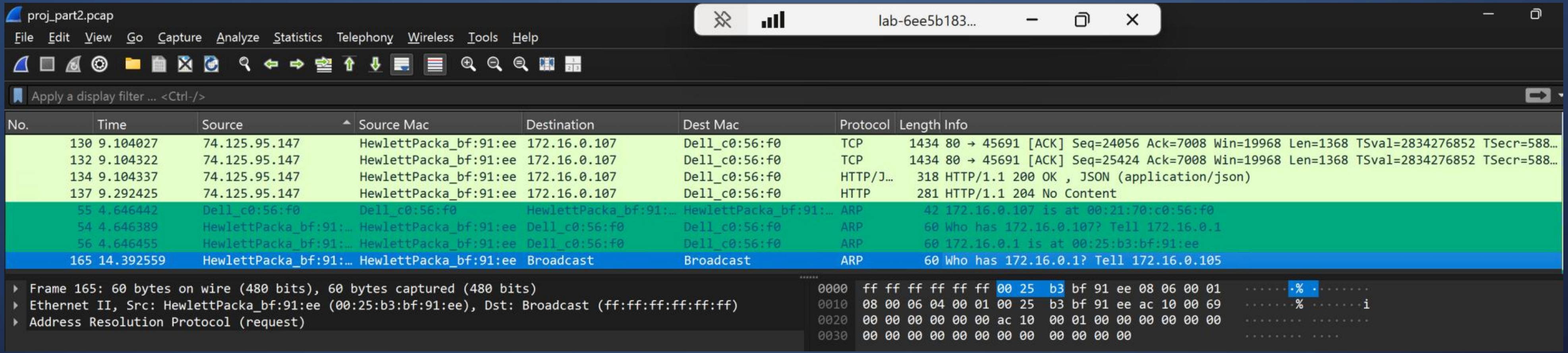


- Packet 57 - MAC address changes to attacker



PACKET ANALYSIS (PACKET 165)

- Packet 165 - ARP request broadcast by attacker.
- Indicates traffic redirection and manipulation.



FINDINGS

- MAC spoofing detected.
- ARP poisoning confirmed.
- Traffic redirected through attacker.
- Persistent malicious ARP activity observed.

CONCLUSION

- Hash validation ensures integrity and reliability of the data analyzed.
- Confirms the presence of a Man-in-the-Middle (MITM) attack using ARP poisoning.
- Victim device communicated with external systems through the router indicates the Normal traffic.
- Attacker positioned between victim and router.
- ARP request is being broadcast by the attacker MAC address to the network.
- Highlights importance of network monitoring and forensic analysis in detecting and responding to security incidents.

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THANK YOU!